

5. Scare You

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Function Description

Working Principle

Experiment Steps

Code Implementation

Method 1: Use the built-in microphone of the micro:bit

Method 2: Use the MIC on the expansion board

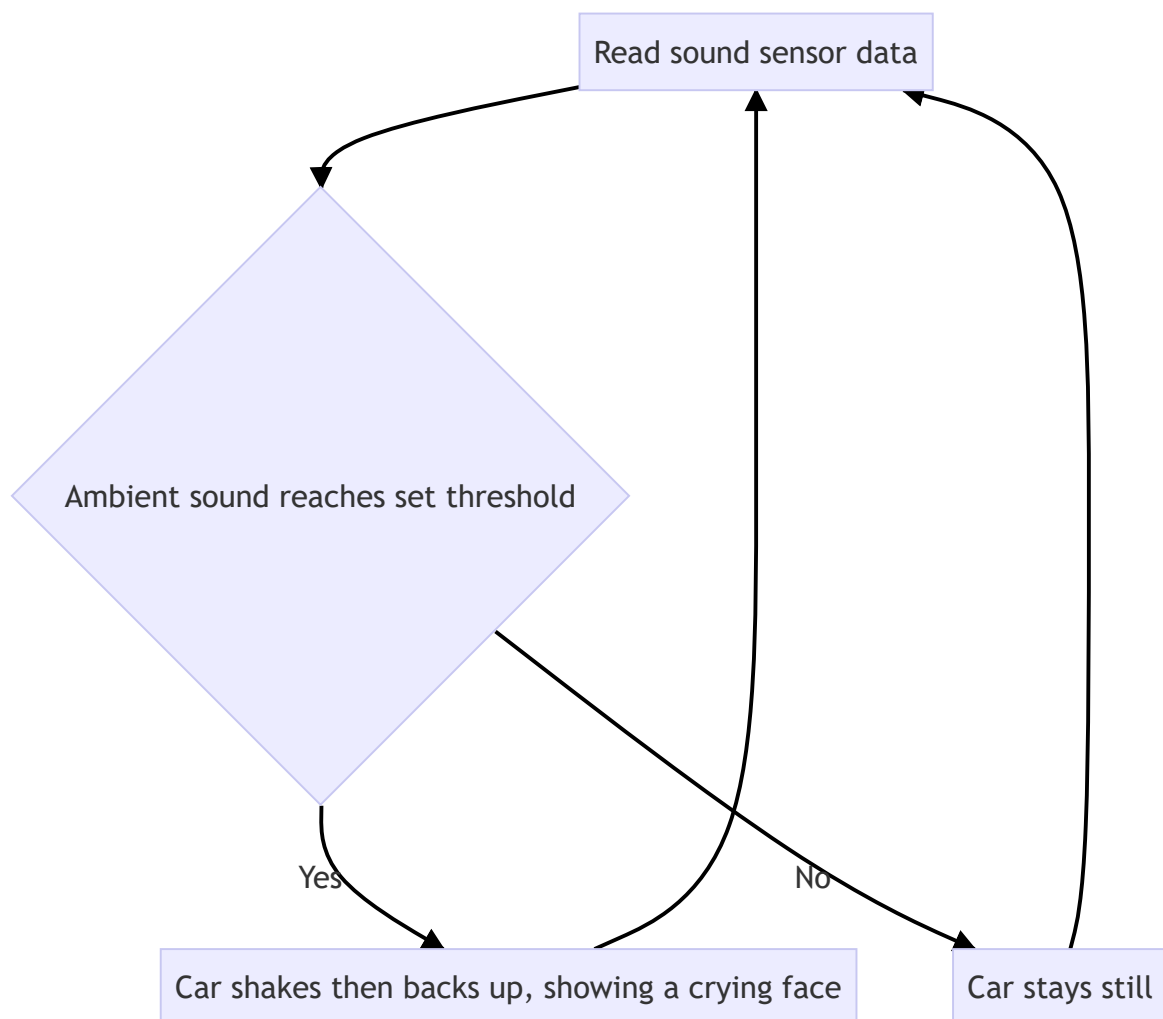
Experimental Phenomenon

Notes

Function Description

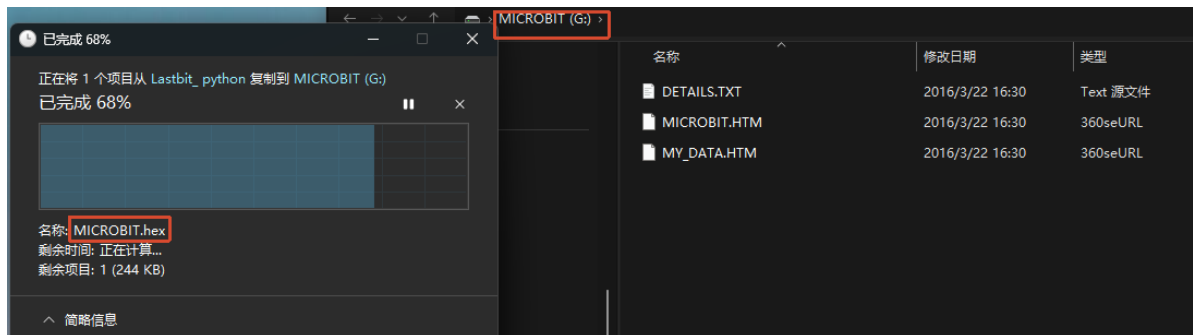
In this section, we will learn to use the microphone sensor to detect sound and achieve the "Scare You" effect. When a loud sound is detected in the environment, the car will show a "scared" reaction: display a sad expression, sway its body, back up, and stop.

Working Principle



Experiment Steps

Make sure `LASTBIT.hex` is flashed before flashing, otherwise an error will occur and the module cannot be imported



1. Before flashing, switch the switch to the OFF position and use a microUSB data cable to connect the computer and the micro:bit main board. **Note that it is the Microbit interface, not the Charging port on the expansion board.**
2. Open the Mu software. Here you can directly copy the program source code we provided; the operation steps are as follows. You can also enter the code manually in the editing window. Note! All English letters and symbols should be entered in English input mode. Use the Tab key for indentation, and the last line should end with a blank program.

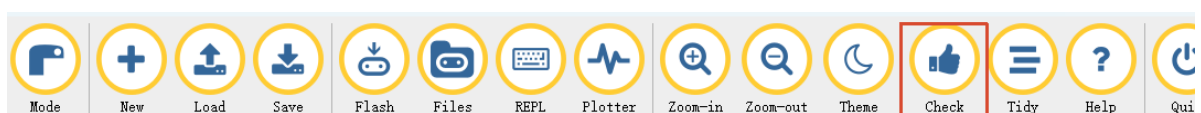
Click the `New` button in the toolbar at the top left, and a file titled "untitled" will appear.



3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a red dot after the title bar, indicating that the code content has been modified but is in an unsaved state.

Whether to save or not does not affect code checking or flashing.

4. Now you can click `Check` to verify whether there are any problems with the code format.

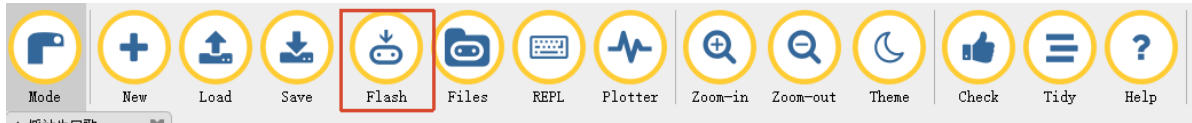


5. Now if you have already connected the MICROBIT to the PC in advance following the steps above, and flashed `LASTBIT.hex`, the next step is to download the current program to the

Microbit.

Click the **Flash** tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently. When the editor shows **Copied code onto micro:bit.** at the bottom, it means the flashing is complete, and the indicator light will no longer blink after flashing.



6. Unplug the microUSB data cable and switch the switch to the ON position.

Code Implementation

Method 1: Use the built-in microphone of the micro:bit

```
# 5. Scare You (using the Microbit Mic)
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

display.show(Image.HEART)

while True:
    # Read sound level
    sound_level = microphone.sound_level()

    # Sound detection threshold can be adjusted
    if sound_level >= 100:
        # Loud sound detected, show sad face and run away
        display.show(Image.SAD)

        # Swing left and right
        for i in range(3):
            Lastbit.set_motor(-60, 60)
            sleep(200)
            Lastbit.set_motor(60, -60)
            sleep(200)

        # Move backward
        Lastbit.car_control('BACKWARD', 200, 300)
        Lastbit.stop_motor()
        sleep(100)
        display.clear()
        sleep(50)
```

Method 2: Use the MIC on the expansion board

```
# 5. Scare You (Expansion board MIC)
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()
```

```

display.show(Image.HEART)

while True:
    # Read sound level (from expansion board MIC)
    sound_level = Lastbit.sound_detect()[2]

    # Sound detection threshold can be adjusted
    if sound_level >= 100:
        # Loud sound detected, show sad face and run away
        display.show(Image.SAD)

        # Swing left and right
        for i in range(3):
            Lastbit.set_motor(-60, 60)
            sleep(200)
            Lastbit.set_motor(60, -60)
            sleep(200)

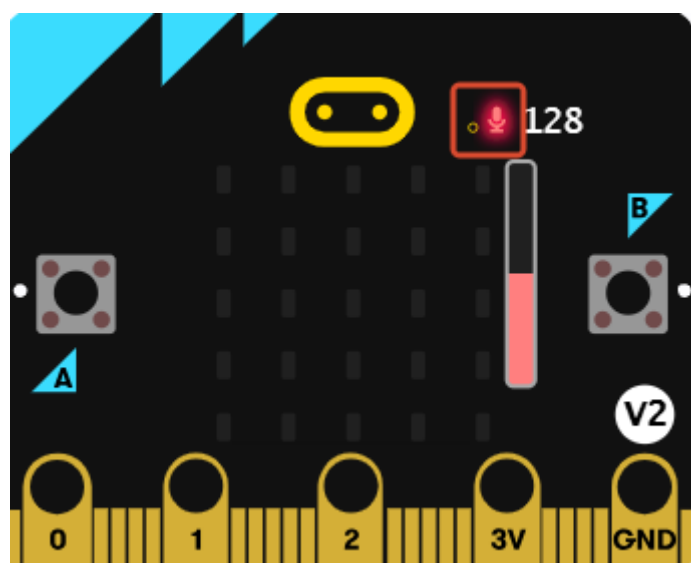
        # Move backward
        Lastbit.car_control('BACKWARD', 200, 300)
        Lastbit.stop_motor()
        sleep(100)
    display.clear()
    sleep(50)

```

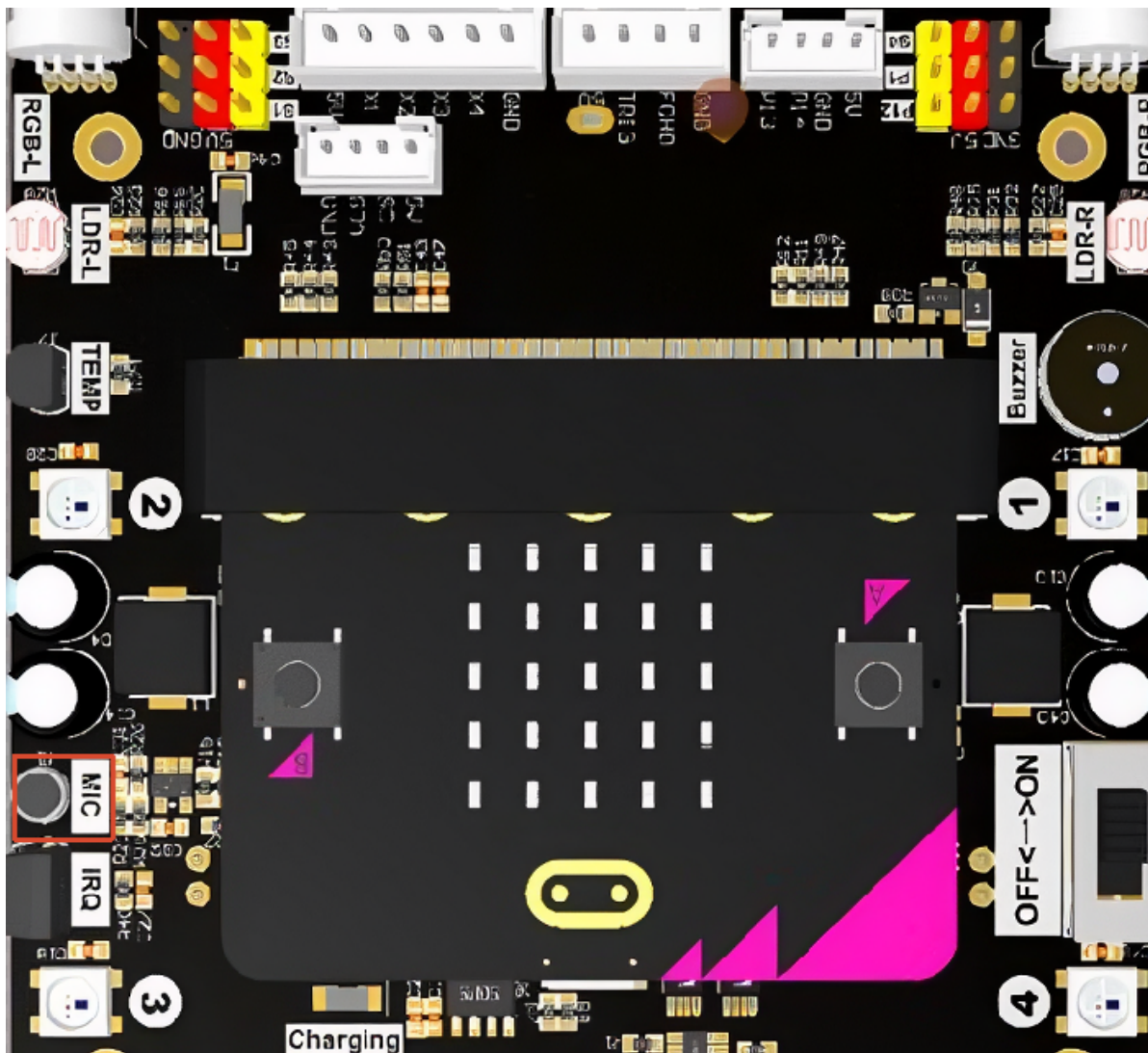
Experimental Phenomenon

The car will continuously detect the ambient sound. When the sound is loud enough, the car will show a crying face, shake, and back up, demonstrating a "scared" effect. Placing the sound source close to the microphone recording position will make it easier to detect the sound. You can also adjust the sound detection threshold according to the actual situation.

The recording position of the built-in microphone of the micro:bit is as follows:



The recording position of the MIC on the expansion board is as follows:



Notes

- Ambient noise may affect the detection effect. It is recommended to conduct the experiment in a relatively quiet environment
- When using the built-in microphone of the micro:bit, use `microphone.sound_level()` to read the sound
- When using the MIC on the expansion board, use `Lastbit.sound_detect()[2]` to read the sound
- The sound detection threshold can be adjusted according to the actual environment